DrainGuard AI

A Flood & Drainage Risk Dashboard for Lagos

WFEO ECBAP AI + Engineering Challenge 2026 — Water, Sanitation & Resilient Infrastructure

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What This Document Is

This is a short, plain explanation of DrainGuard AI: what it does, how it works under the hood, and what is real data versus what is simulated. It's written so it can be handed to anyone — a judge, a teammate, or a curious friend — and understood in a few minutes.

1 The Problem, In One Paragraph

Lagos floods often and badly. A big reason is that drains get blocked or overwhelmed, and nobody finds out until water is already in the street. There's no simple, live picture of *which drains, in which neighborhoods, are most at risk right now.* DrainGuard AI is an attempt to build that picture cheaply, using real weather data and honest engineering, without needing expensive hardware nobody has yet installed.

2 What DrainGuard AI Does

DrainGuard AI is a web dashboard that shows, for a set of real flood-prone locations across Lagos:

- An overall flood risk level for the city (Low / Moderate / High), shown as a gauge.
- A live map with a pin at each monitored drain, color-coded by risk.
- A table of every drain's water level, blockage percentage, and risk score.
- Real rainfall, temperature, and weather conditions for Lagos, refreshed automatically.
- A rainfall trend chart and a recent-alerts feed.

3 How It Works (Technical View)

Stack: Next.js (frontend, React) + Convex (backend: database, scheduled jobs, serverless functions). No authentication layer — the dashboard is public/read-only, so nothing is gated behind login.

Data flow:

1. Every 7 minutes, a Convex cron job calls the Open-Meteo weather API (free, no key required) and pulls real 24-hour rainfall, current temperature, weather condition, and soil moisture for Lagos.
2. That same cron cycle updates a simulated reading (water level, blockage %) for each of the 10 monitored drains. The simulation is a random walk nudged by the real rainfall figure — heavier real rain pushes simulated water levels up, exactly like it would in reality.
3. A scoring function combines water level, blockage, and rainfall into a single 0–100 **risk score** per drain, and an overall city-wide score.
4. The frontend subscribes to this data live (Convex pushes updates instantly to the browser — no manual refresh needed).

Risk score formula (transparent, not a black box):

$$Score = \underbrace{40 \times waterlevel 150cm}_{water} + \underbrace{30 \times blockage\%100}_{blockage} + \underbrace{30 \times rainfall(24h)50mm}_{rainfall}$$

Each term is capped so it can't exceed its share. Score ≥ 70 = High, ≥ 40 = Moderate, else Low. The 150cm and 50mm reference values are reasonable placeholders, not site-surveyed engineering constants — flagged honestly in the code for future calibration.

4 What's Real, and What's Simulated

Being honest about this is the whole point of an MVP. Nobody has physical flood sensors installed in Lagos drains at this scale — not us, not (currently) anyone outside a small pilot. So:

- **Real:** the 10 drain locations (actual named, flood-prone Lagos neighborhoods), all rainfall/temperature/weather/soil-moisture data (live from Open-Meteo), and the risk-scoring math applied to that data.
- **Simulated:** water level and blockage percentage per drain, because no live sensor exists yet. The simulation is not random noise — it responds to real rainfall, so it behaves plausibly, but it is not a live sensor reading and the dashboard doesn't pretend otherwise.
- **Built but not yet active:** a community-reporting feature, where residents could report a flooded drain and have it feed into the real data (a zero-cost path to genuine sensor-free ground truth). The backend for this exists; the submission form in the UI is still being built.

Nothing on the dashboard is hardcoded or faked to look more real than it is. Where a genuine data source wasn't available (e.g. tide level), the feature was removed rather than invented.

5 Why This Approach, Not Fake Sensors

It would have been easy to just make up sensor numbers and call it "AI-powered." We chose not to, because a dashboard that quietly fabricates data is worse than one that's honest about being a prototype. Every number that *can* be real (weather, location, the math) *is* real. Every number that can't yet be real (physical sensor readings) is clearly built on a transparent simulation, not disguised as one.

6 In One Sentence

DrainGuard AI turns live Lagos weather data into a real-time, explainable flood-risk picture for ten real drainage hotspots — honestly built as a working prototype, not a finished sensor network.

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